

3. Homomorphism and isomorphism

3.1. HOMOMORPHISM AND ISOMORPHISM

READING. [moore2023].

Chapter 2 produced several groups that were visibly “the same”: reflection and particle exchange both gave $\mathbb{Z}_2$; $SO(2)$ and $U(1)$ had identical group laws; D_2 , V_4 and $\mathbb{Z}_2 \times \mathbb{Z}_2$ kept reappearing. This chapter makes “the same” precise.

Definition 3.1 (Homomorphism). Let $(G, \mathbf{m}, \mathbf{I}, e)$ and $(G', \mathbf{m}', \mathbf{I}', e')$ be groups. A map $\varphi : G \rightarrow G'$ is a *homomorphism* if

homomorphism
同态

$$\varphi(\mathbf{m}(g_1, g_2)) = \mathbf{m}'(\varphi(g_1), \varphi(g_2)), \quad \text{i.e.} \quad \varphi(g_1 g_2) = \varphi(g_1) \varphi(g_2)$$

for all $g_1, g_2 \in G$.

Equivalently, the square

$$\begin{array}{ccc} G \times G & \xrightarrow{\mathbf{m}} & G \\ \varphi \times \varphi \downarrow & & \downarrow \varphi \\ G' \times G' & \xrightarrow{\mathbf{m}'} & G' \end{array}$$

commutes.

commutative diagram
交换图表

A homomorphism automatically respects the rest of the structure: From $\varphi(e) = \varphi(e \cdot e) = \varphi(e)\varphi(e)$ we get $\varphi(e) = e'$. Then $e' = \varphi(e) = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1})$, so $\varphi(g^{-1}) = [\varphi(g)]^{-1}$.

Remark 3.2 (Injective, surjective, and the special cases).

1. φ is *injective* if and only if $\varphi(g) = e' \Rightarrow g = e$.

Indeed, $\varphi(g_1) = \varphi(g_2)$ gives $e' = \varphi(g_1)\varphi(g_2)^{-1} = \varphi(g_1g_2^{-1})$, so $g_1g_2^{-1} = e$ and $g_1 = g_2$.

2. φ is *surjective* if for every $g' \in G'$ there is $g \in G$ with $\varphi(g) = g'$.

3. φ is an *isomorphism* if it is both, i.e. bijective. Then φ^{-1} is also a homomorphism, since

isomorphism
同构
bijective
双射

$$\varphi^{-1}(g'_1 g'_2) = \varphi^{-1}(\varphi(g_1)\varphi(g_2)) = g_1 g_2 = \varphi^{-1}(g'_1) \varphi^{-1}(g'_2).$$

Isomorphism is an equivalence relation, and isomorphic groups are regarded as the same group.

automorphism
自同构

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4. An isomorphism $\varphi : G \rightarrow G$ of a group with itself is an *automorphism*.

Example 3.3 (Automorphisms of $\mathbb{Z}_N$). Since $\mu_4 \cong \mathbb{Z}_4$, consider maps $\mathbb{Z}_N \rightarrow \mathbb{Z}_N$ of the form $\bar{x} \mapsto k\bar{x}$. This is an automorphism precisely when

$$\gcd(k, N) = 1,$$

since otherwise the map fails to be injective. For $N = 4$ the choice $k = 3$ works, whereas $k = 2$ does not.

KEY POINTS

- A homomorphism is a map that respects multiplication; respecting the identity and inverses then comes for free.
- Injectivity can be tested at the identity alone, $\varphi(g) = e' \Rightarrow g = e$, which is what makes §1.2 work.
- An isomorphism is a bijective homomorphism; isomorphic groups are “the same group”, which is what lets us say $SO(2) \cong U(1)$.
- $\bar{x} \mapsto k\bar{x}$ is an automorphism of $\mathbb{Z}_N$ iff $\gcd(k, N) = 1$.

3.2. KERNEL AND IMAGE

READING. [moore2023].

A homomorphism can fail to be injective, and it can fail to be surjective. Two subgroups measure the two failures.

kernel
核
image
像

Definition 3.4 (Kernel and image). Let $\varphi : G \rightarrow G'$ be a homomorphism. Its *kernel* and *image* are

$$\ker \varphi := \{ g \in G : \varphi(g) = e' \} \subset G, \quad \text{im } \varphi := \varphi(G) \subset G'.$$

Proposition 3.5. $\ker \varphi$ is a subgroup of G , and $\text{im } \varphi$ is a subgroup of G' .

Proof. Both contain the identity, by $\varphi(e) = e'$, so both are nonempty. If $g_1, g_2 \in \ker \varphi$ then

$$\varphi(g_1 g_2^{-1}) = \varphi(g_1) \varphi(g_2)^{-1} = e',$$

so $g_1 g_2^{-1} \in \ker \varphi$. If $g'_1 = \varphi(g_1)$ and $g'_2 = \varphi(g_2)$ lie in $\text{im } \varphi$ then $g'_1 (g'_2)^{-1} = \varphi(g_1 g_2^{-1}) \in \text{im } \varphi$. Apply Lemma 2.5 to each. $\square$

Proposition 3.6. φ is injective if and only if $\ker \varphi = \{e\}$. Hence φ is an isomorphism if and only if

$$\ker \varphi = \{e\} \quad \text{and} \quad \text{im } \varphi = G'.$$

Proof. The first statement is Remark 1.2(1) written in the present notation, and the second combines it with the definition of $\text{im } \varphi$. $\square$

Quiz 3.1. Show that $\ker \varphi$ absorbs conjugation, that is $gkg^{-1} \in \ker \varphi$ for every $k \in \ker \varphi$ and every $g \in G$. Subgroups with this property are the subject of Chapter 7.

3.3. EXAMPLES

Example 3.7 ($\mathbb{Z}_N \cong \mu_N$). Define $\varphi : \mathbb{Z}_N \rightarrow \mu_N$ by $\varphi(\bar{r}) = e^{2\pi i r/N}$. Replacing r by $r + N$ does not change the value, so φ is well defined on residue classes, and

$$\varphi(\bar{r}_1 + \bar{r}_2) = e^{2\pi i(r_1 + r_2)/N} = \varphi(\bar{r}_1) \varphi(\bar{r}_2)$$

makes it a homomorphism from addition to multiplication. Its kernel is $\{\bar{0}\}$ and its image is all of μ_N , so $\mathbb{Z}_N \cong \mu_N$ by Proposition 1.6. The residue classes of §2.3 and the roots of unity are the same group written twice.

Example 3.8 (The power map). For $k \in \mathbb{Z}$ let $P_k : \mu_N \rightarrow \mu_N$ be $z \mapsto z^k$. It is a homomorphism, since $(z_1 z_2)^k = z_1^k z_2^k$. Its kernel is

$$\ker P_k = \{z \in \mu_N : z^k = 1\} = \mu_d, \quad d = \gcd(k, N),$$

so P_k is an isomorphism exactly when $\gcd(k, N) = 1$, which is Example 1.3 again in multiplicative notation. At the other extreme P_N is the trivial map. For $N = 4$ and $k = 2$,

$$\mu_4 = \{1, i, -1, -i\}, \quad \ker P_2 = \{\pm 1\} \cong \mathbb{Z}_2, \quad \text{im } P_2 = \{\pm 1\} \cong \mu_2.$$

Example 3.9 ($U(1) \rightarrow SU(2)$). Fix an integer $N \geq 2$ and send $z \in U(1)$ to

$$\varphi(z) = \begin{pmatrix} z^N & 0 \\ 0 & z^{-N} \end{pmatrix}.$$

The matrix is unitary with determinant 1, so φ maps into $SU(2)$, and $\varphi(z_1 z_2) = \varphi(z_1) \varphi(z_2)$. Here

$$\ker \varphi = \{z : z^N = 1\} = \mu_N, \quad \text{im } \varphi = \left\{ \begin{pmatrix} w & 0 \\ 0 & w^{-1} \end{pmatrix} : |w| = 1 \right\} \cong U(1),$$

so this φ is neither injective nor surjective, and the two subgroups say by how much it misses.

READING. [moore2023].

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3.4. $SU(2)$ AND $SO(3)$

A rotation acts on a vector by an element of $SO(3)$ and on a $\text{spin-}\frac{1}{2}$ state by an element of $SU(2)$. The two are related by a homomorphism, which we now build.

Let $\mathcal{H}$ be the set of traceless Hermitian 2×2 matrices, a three-dimensional real vector space. Send a vector to such a matrix by

$$h : \mathbb{R}^3 \rightarrow \mathcal{H}, \quad h(\vec{x}) = \vec{x} \cdot \vec{\sigma} = x_i \sigma^i = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}.$$

This is a linear bijection, and the length of $\vec{x}$ is stored in the determinant,

$$\det h(\vec{x}) = -x_3^2 - (x_1^2 + x_2^2) = -|\vec{x}|^2.$$

For $u \in SU(2)$ let $C_u(m) = u m u^\dagger$. Then C_u maps $\mathcal{H}$ to itself, since $\text{Tr}(umu^\dagger) = \text{Tr}(m) = 0$ and $(umu^\dagger)^\dagger = umu^\dagger$.

Definition 3.10 (The rotation attached to u). $R(u) := h^{-1} \circ C_u \circ h$, that is, $R(u)$ is the map $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ determined by

$$u(\vec{x} \cdot \vec{\sigma})u^\dagger = (R(u)\vec{x}) \cdot \vec{\sigma} \quad \text{for all } \vec{x} \in \mathbb{R}^3, \quad \text{equivalently} \quad u \sigma^i u^\dagger = R(u)_{ji} \sigma^j.$$

Theorem 3.11. $R : SU(2) \rightarrow SO(3)$ is a surjective homomorphism with $\ker R = \{\pm 1\} \cong \mathbb{Z}_2$.

Proof. Homomorphism. Conjugating twice,

$$(R(u_1 u_2) \vec{x}) \cdot \vec{\sigma} = (u_1 u_2)(\vec{x} \cdot \vec{\sigma})(u_1 u_2)^\dagger = u_1 [(R(u_2) \vec{x}) \cdot \vec{\sigma}] u_1^\dagger = (R(u_1) R(u_2) \vec{x}) \cdot \vec{\sigma},$$

and h is injective, so $R(u_1 u_2) = R(u_1) R(u_2)$.

Orthogonal. Conjugation does not change a determinant, so

$$|R(u)\vec{x}|^2 = -\det((R(u)\vec{x}) \cdot \vec{\sigma}) = -\det(u(\vec{x} \cdot \vec{\sigma})u^\dagger) = -\det(\vec{x} \cdot \vec{\sigma}) = |\vec{x}|^2,$$

and $R(u) \in O(3)$.

Determinant. Write the product rule with a summed index ℓ , as $\sigma^i \sigma^j = \delta^{ij} \mathbb{1} + i\epsilon^{ij\ell} \sigma^\ell$, multiply on the right by σ^k , and take the trace. With $\text{Tr} \sigma^k = 0$ and $\text{Tr}(\sigma^\ell \sigma^k) = 2\delta^{\ell k}$,

$$\text{Tr}(\sigma^i \sigma^j \sigma^k) = 2i \epsilon^{ijk}.$$

The factors $u^\dagger u$ inserted between the three matrices cancel under the trace, so $\text{Tr}(\sigma^1 \sigma^2 \sigma^3) = \text{Tr}((u\sigma^1 u^\dagger)(u\sigma^2 u^\dagger)(u\sigma^3 u^\dagger))$. Substituting $u\sigma^a u^\dagger = R(u)_{ia} \sigma^i$,

$$\begin{aligned} 2i &= \text{Tr}(\sigma^1 \sigma^2 \sigma^3) = R(u)_{i1} R(u)_{j2} R(u)_{k3} \text{Tr}(\sigma^i \sigma^j \sigma^k) \\ &= 2i \epsilon^{ijk} R(u)_{i1} R(u)_{j2} R(u)_{k3} = 2i \det R(u). \end{aligned}$$

So $\det R(u) = 1$ and $R(u) \in SO(3)$.

Kernel. $R(u) = \mathbb{1}$ says $u\sigma^i u^\dagger = \sigma^i$ for each i , so u commutes with $\sigma^1, \sigma^2, \sigma^3$ and therefore with every 2×2 matrix, those three together with $\mathbb{1}$ being a basis. By Proposition 2.17, $u = \lambda \mathbb{1}$, and $\det u = \lambda^2 = 1$ forces $\lambda = \pm 1$.

Surjective. Take $u = \exp(-\frac{i}{2}\theta\sigma^3) = \text{diag}(e^{-i\theta/2}, e^{i\theta/2})$. Then

$$u\sigma^1 u^\dagger = \begin{pmatrix} 0 & e^{-i\theta} \\ e^{i\theta} & 0 \end{pmatrix} = \cos\theta\sigma^1 + \sin\theta\sigma^2, \quad u\sigma^3 u^\dagger = \sigma^3,$$

so $R(u)$ is the rotation by θ about the third axis. Rotations about the other two axes come the same way from σ^1 and σ^2 , and these generate $SO(3)$. $\square$

Remark 3.12 (Double cover). The two minus signs cancel in $um u^\dagger$, so $R(u) = R(-u)$ and R is two-to-one. Since $(\hat{n} \cdot \vec{\sigma})^2 = \mathbb{1}$ for a unit vector $\hat{n}$, expanding the exponential gives

$$u(\hat{n}, \theta) = \exp\left(-\frac{i}{2}\theta \hat{n} \cdot \vec{\sigma}\right) = \cos\frac{\theta}{2} \mathbb{1} - i \sin\frac{\theta}{2} \hat{n} \cdot \vec{\sigma},$$

so $u(\hat{n}, 2\pi) = -\mathbb{1}$ and $u(\hat{n}, 4\pi) = \mathbb{1}$. A rotation by 2π returns every vector to where it started but multiplies a spin- $\frac{1}{2}$ wavefunction by -1 ; at 4π the wavefunction returns to its original value.

Geometrically, $SU(2) \sim S^3$ by Example 2.20, and R identifies the antipodal pair $\pm u$. So $SO(3)$ is the three-sphere with antipodes identified, the real projective space $\mathbb{RP}^3$.

3.5. $GL(V)$ AND $GL(n, \kappa)$

Example 3.13 (Coordinates as an isomorphism). Let V be an n -dimensional vector space over κ and let $GL(V)$ be the group of invertible linear maps $V \rightarrow V$. An ordered basis $b = \{\hat{e}_1, \dots, \hat{e}_n\}$ turns each $\tau \in GL(V)$ into a matrix $T_b(\tau)$ through

$$\tau(\hat{e}_i) = \sum_j \hat{e}_j T_b(\tau)_{ji}.$$

Applying τ_1 after τ_2 to $\vec{v} = \sum_i v_i \hat{e}_i$,

$$\tau_1(\tau_2 \vec{v}) = \sum_{i,j} v_i T_b(\tau_2)_{ji} \tau_1(\hat{e}_j) = \sum_{i,j,k} \hat{e}_k T_b(\tau_1)_{kj} T_b(\tau_2)_{ji} v_i,$$

which is $\sum_{i,k} \hat{e}_k T_b(\tau_1 \tau_2)_{ki} v_i$, so

$$T_b(\tau_1 \tau_2) = T_b(\tau_1) T_b(\tau_2)$$

and $\varphi_b : \tau \mapsto T_b(\tau)$ is a homomorphism $GL(V) \rightarrow GL(n, \kappa)$. It is surjective, since every invertible matrix defines an invertible map through the same formula. It is injective, since $T_b(\tau) = \mathbb{1}_n$ says $\tau(\hat{e}_i) = \hat{e}_i$ for every i and hence $\tau = \text{id}$. So

$$GL(V) \cong GL(n, \kappa).$$

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The isomorphism depends on the basis. A second basis $\hat{e}_i = \sum_j S_{ji} \hat{e}'_j$ gives $T_{b'}(\tau) = S T_b(\tau) S^{-1}$, so the two coordinate pictures differ by conjugation. This is why an abstract group can be studied through matrices at all, and it is where representation theory starts.

KEY POINTS

- $\ker \varphi$ and $\text{im } \varphi$ are subgroups, and they measure the failure of φ to be injective and surjective. φ is an isomorphism exactly when $\ker \varphi = \{e\}$ and $\text{im } \varphi = G'$.
- $R : SU(2) \rightarrow SO(3)$ is onto with kernel $\{\pm 1\}$, so $SU(2)$ is a double cover of $SO(3)$ and $SO(3) \sim \mathbb{RP}^3$.
- A rotation by 2π is the identity on vectors and -1 on spin- $\frac{1}{2}$ states.
- Choosing a basis gives $GL(V) \cong GL(n, \kappa)$; changing the basis conjugates the matrices.

KEY TERMS. homomorphism 同态 · isomorphism 同构 · automorphism 自同构
· injective 单射 · surjective 满射 · bijective 双射 · kernel 核 · image 像 · double cover 双覆盖

